5.8 Integration

5.8.1 Algebras of Sets

Definition 5.8.1
Let Ω be a set and $\mathcal{F}$ a family of subsets of Ω. We consider the following types of families

*a) $\mathcal{F}$ is a **semi-ring** if the following properties hold*
- *i) $\emptyset \in \mathcal{F}$*
- *ii) $A_1, ..., A_n \in \mathcal{F} \implies A_1 \cap ... \cap A_n \in \mathcal{F}$*
- *iii) For all $A, B \in \mathcal{F}$ there is a family of disjoint subsets $C_1, ..., C_n \in \mathcal{F}$ such that*

$$B \setminus A = C_1 \sqcup ... \sqcup C_n$$

*b) $\mathcal{F}$ is a **ring** (resp. **algebra**) if the following properties hold*
- *i) $\emptyset \in \mathcal{F}$ (resp. $\Omega \in \mathcal{F}$)*
- *ii) $A_1, ..., A_n \in \mathcal{F} \implies A_1 \cup ... \cup A_n \in \mathcal{F}$*
- *iii) $A, B \in \mathcal{F} \implies A \setminus B \in \mathcal{F}$*

*c) $\mathcal{F}$ is a **σ-ring** (resp. **σ-algebra**) if the following properties hold*
- *i) $\emptyset \in \mathcal{F}$ (resp. $\Omega \in \mathcal{F}$)*
- *ii) $A_1, ..., A_n, ... \in \mathcal{F} \implies \bigcup_{i=1}^{\infty} A_i \in \mathcal{F}$*
- *iii) $A, B \in \mathcal{F} \implies A \setminus B \in \mathcal{F}$*

Evidently σ-ring $\implies$ ring $\implies$ semi-ring.

Example 5.8.2
The family of half-open intervals $\mathcal{F} = \{[a, b) \mid a < b\}$ forms a semi-ring on $\mathbb{R}$.

More generally the family of products of half-open intervals $\{[a_1, b_1) \times ... \times [a_n, b_n) \mid a_i < b_i\}$ forms a semi-ring on $\mathbb{R}^n$.

Lemma 5.8.3
Let Ω be a set and $\{\mathcal{F}_\alpha\}_{\alpha \in \mathcal{A}}$ a family of rings (resp. algebras, σ-rings, σ-algebras). Then the family of subsets

$$\bigcap_{\alpha \in \mathcal{A}} \mathcal{F}_\alpha$$

is a ring (resp. algebra, σ-ring, σ-algebra). In particular every family of subsets is contained in a smallest ring (resp. algebra, σ-ring, σ-algebra).

Definition 5.8.4 (Borel Algebra)
*Let X be a topological space. We say that the family of **borel sets** $\mathcal{B}(X)$ is the smallest σ-algebra containing the family of open sets.*

Proposition 5.8.5
Let $\mathcal{R}$ be a semi-ring (resp. semi-algebra). Then the family of finite disjoint unions

$$\mathcal{F} := \{A_1 \sqcup ... \sqcup A_n \mid A_1, ..., A_n \in \mathcal{R} \text{ pairwise disjoint}\}$$

is a ring (resp. algebra), and indeed the smallest ring (resp. algebra) containing $\mathcal{R}$.

Lemma 5.8.6 (Semi-Ring Extended Set Difference)
Let $\mathcal{R}$ be a semi-ring and $A_0, A_1, ..., A_n \in \mathcal{R}$. Then

$$A_0 \setminus \bigcup_{i=1}^{n} A_i = \bigsqcup_{k=1}^{m} B_k$$

for some disjoint sets $B_k \in \mathcal{R}$.

Proof. We proceed by induction, the case $n = 1$ holding by definition. Suppose the relation holds for n. Then

$$A_0 \setminus \bigcup_{i=1}^{n+1} A_i = \left(\bigsqcup_{k=1}^{m} B_k \right) \setminus A_{n+1} = \bigsqcup_{k=1}^{m} (B_k \setminus A_{n+1})$$

and by definition of $\mathcal{R}$ we have

$$B_k \setminus A_{n+1} = \bigsqcup_{j=1}^{N} C_{kj} \quad k = 1...m$$

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and $C_{kj} \in \mathcal{R}$. As B_k are disjoint we see that the C_{kj} are disjoint for all j,k and the result follows immediately. ■

5.8.2 Set Functions

Definition 5.8.7 (Finitely Additive Set Function)
Let $\mathcal{F}$ be a family of subsets of Ω containing $\emptyset$. A ***set function***

$$\mu : \mathcal{F} \to [0, \infty]$$

is ***finitely additive*** *if it satisfies*

a) $\mu(\emptyset) = 0$

b) $\mu(A_1 \sqcup ... \sqcup A_n) = \sum_{i=1}^{n} \mu(A_i)$ *whenever this is well-defined*

We say a set $A \in \mathcal{F}$ is ***finite*** *if $\mu(A) < \infty$, and μ is* ***finite*** *if $\mu(\mathcal{F}) < \infty$.*

*We say that μ is σ-****finite*** *if for every set $A \in \mathcal{F}$ there exists a sequence A_i such that $A \subset \bigcup_{i=1}^{\infty} A_i$ and $\mu(A_i) < \infty$. If $\Omega \in \mathcal{F}$ then it is sufficient for this condition to hold for Ω.*

Proposition 5.8.8 (Extension Theorem I)
Let $\mathcal{R}$ be a semi-ring, and $\mathcal{F}$ the ring generated by $\mathcal{R}$. Given a finitely additive set function $\mu : \mathcal{R} \to [0, \infty]$ there exists a unique extension to $\mathcal{F}$ given by

$$\mu(A_1 \sqcup ... \sqcup A_n) = \sum_{i=1}^{n} \mu(A_i)$$

which is finitely additive.

Proof. In order to ensure the definition is well-defined suppose

$$A_1 \sqcup ... \sqcup A_n = B_1 \sqcup ... \sqcup B_m$$

Then in particular

$$A_i = (A_i \cap B_1) \sqcup ... \sqcup (A_i \cap B_m)$$

We may deduce by the additive property that

$$\sum_{i=1}^{n} \mu(A_i) = \sum_{i=1}^{n} \sum_{j=1}^{m} \mu\big(A_i \cap B_j\big)$$

Evidently this is symmetric in A and B so this also equals $\sum_{j=1}^{m} \mu\big(B_j\big)$. This shows that the measure μ is well-defined. The additivity property is straightforward. ■

Proposition 5.8.9 (Properties of finitely-additive set functions)
Let $(\Omega, \mathcal{R})$ be a ***semi-ring*** *and $\mu : \mathcal{R} \to [0, \infty]$ a finitely additive set function. Then*

a) μ *is* ***monotone****, that is for sets $A \subset B$ we have*

$$\mu(A) \leq \mu(B)$$

b) Suppose $A_1 \sqcup ... \sqcup A_n \subset A_0$ for $A_i \in \mathcal{R}$ then

$$\sum_{i=1}^{n} \mu(A_i) \leq \mu(A_0)$$

c) Suppose $A_0 \subset A_1 \cup ... \cup A_n$ for $A_i \in \mathcal{R}$ then

$$\mu(A_0) \leq \sum_{i=1}^{n} \mu(A_i)$$

Proof.

a) By definition $B \setminus A = C_1 \sqcup ... \sqcup C_n$ and therefore by finite additivity

$$\mu(B) = \mu(C_1 \sqcup ... \sqcup C_n \sqcup A) = \mu(C_1) + ... + \mu(C_n) + \mu(A) \geq \mu(A)$$

b) By (5.8.6)

$$A_0 \setminus \bigsqcup_{i=1}^{n} A_i = \bigsqcup_{k=1}^{m} B_k$$

for some $B_k \in \mathcal{R}$ disjoint. Observe they are by construction disjoint from A_i therefore

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$$A_0 = A_1 \sqcup ... \sqcup A_n \sqcup B_1 \sqcup ... \sqcup B_m$$

The result follows from finite additivity.

c) By replacing A_i with $A_i \cap A_0$ we may assume that $A_0 = A_1 \cup ... \cup A_n$ by a). Observe that by (5.8.6)

$$\hat{A}_i := A_i \setminus \bigcup_{j=1}^{i-1} A_j = \bigsqcup_{k=1}^{m} C_{ik}$$

for $C_{ik} \in \mathcal{R}$ (if necessary padding out by $\emptyset$). Therefore

$$A_0 = \bigsqcup_{i=1}^{n} \hat{A}_i = \bigsqcup_{i,k} C_{ik}$$

and by finite additivity

$$\mu(A_0) = \sum_{i=1}^{n} \sum_{k=1}^{m} \mu(C_{ik})$$

By b)

$$\sum_{k=1}^{m} \mu(C_{ik}) \leq \mu(A_i)$$

from which the result follows.

■

Definition 5.8.10 (Simple Functions)
Let $(\Omega, \mathcal{F})$ be a family of subsets and X a normed vector space over $\mathbb{R}$. We define the space of simple functions to be

$$\text{Simple}(\mathcal{F}, X) := \left\{ \sum_{i=1}^{n} \lambda_i \mathbf{1}_{A_i} \;\middle|\; A_1 \sqcup ... \sqcup A_n = \Omega, \lambda_1, ..., \lambda_n \in X \right\}$$

Proposition 5.8.11 (Integration of Simple Functions)
Let $(\Omega, \mathcal{F})$ be a ring and μ a finitely additive measure and X a vector space over $\mathbb{R}$. There is a well-defined linear map

$$\int \mathrm{d}\mu : \text{Simple}(\mathcal{F}, X) \to X$$

$$\sum_{i=1}^{n} \alpha_i \mathbf{1}_{A_i} \to \sum_{i=1}^{n} \alpha_i \mu(A_i)$$

Proof. Suppose that $\sum_{i=1}^{n} \alpha_i \mathbf{1}_{A_i} = \sum_{j=1}^{m} \beta_j \mathbf{1}_{B_j}$ where both $\{A_i\}$ and $\{B_j\}$ are *disjoint* partitions of X. Then

$$A_i = (A_i \cap B_1) \sqcup ... \sqcup (A_i \cap B_m)$$

whence

$$\mu(A_i) = \sum_{j=1}^{m} \mu(A_i \cap B_j)$$

Therefore

$$\sum_{i=1}^{n} \alpha_i \mu(A_i) = \sum_{i=1}^{n} \sum_{j=1}^{m} \alpha_i \mu(A_i \cap B_j)$$

By the same argument

$$\sum_{j=1}^{m} \beta_j \mu(B_j) = \sum_{i=1}^{n} \sum_{j=1}^{m} \beta_j \mu(A_i \cap B_j)$$

Evidently $\{A_i \cap B_j\}$ is a disjoint partition of X. If $A_i \cap B_j \neq \emptyset$ then the first equality requires that $\alpha_i = \beta_j$. This shows that the integrals are equal as required. ■

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Proposition 5.8.12
Let $(\Omega, \mathcal{F})$ be a ring, μ a finitely additive measure and X a normed vector space over $\mathbb{R}$. Then we have the following properties

a) $\int (f+g)\,\mathrm{d}\mu = \int f\,\mathrm{d}\mu + \int g\,\mathrm{d}\mu$
b) $\int \lambda f\,\mathrm{d}\mu = \lambda \int f\,\mathrm{d}\mu$
c) $\left\|\int f\,\mathrm{d}\mu\right\| \leq \int \|f\|\,\mathrm{d}\mu$

If μ is finite then

$$\left\|\int f\,\mathrm{d}\mu\right\| \leq \mu(\Omega)\|f\|_\infty$$

Proof. For the last part observe that

$$\left\|\sum_{i=1}^{n} \alpha \mathbf{1}_{A_i}\right\| \leq \sum_{i=1}^{n} \|\alpha\|_i \mu(A_i) \leq \|f\|_\infty \sum_{i=1}^{n} \mu(A_i) \leq \mu(\Omega)\|f\|_\infty$$

■

5.8.3 Integration of Regulated Functions

Definition 5.8.13 (SubBorel Measure)
Let $\Omega = \mathbb{R}$ and $\mathcal{F}$ the ring generated by the half-open subintervals. Then there is a finitely additive measure $\mu : \mathcal{F} \to \mathbb{R}$ determined by

$$\mu([a,b)) = (b-a)$$

By (...) this determines a bounded linear map

$$\int : \mathrm{Simple}(\mathbb{R}, X) \to X$$

for any real vector space X.

Definition 5.8.14 (Regulated Function)
*Let X be a Banach space. A **regulated function** is a map*

$$f : \mathbb{R} \to X$$

*of **compact** support for which there exists a sequence (f_n) of* simple *functions (with respect to the ring generated by the half-open subintervals) such that*

$$f_n \to f$$

converges uniformly. We denote this by

$$\mathrm{Reg}(\mathbb{R}, X)$$

Lemma 5.8.15
A uniform limit of bounded functions is bounded. In particular a regulated function is bounded.

Proof. There is some N such that $n \geq N \implies \|f_n(x) - f(x)\| < 1$. Then

$$\|f(x)\| \leq \|f_n(x) - f(x)\| + \|f_n(x)\| \leq 1 + \|f_n(x)\|$$

We may take supremum over x to show that f is bounded. ■

Proposition 5.8.16 (Continuous Functions are Regulated)
Let X be a Banach space and $f : [a,b] \to X$ a continuous function. Then f is regulated.

Proof. We may define

$$f_n(x) := \sum_{i=1}^{N} f(a_i)\mathbf{1}_{[a_i,b_i)}(x) + \mathbf{1}\{x = b\}f(b)$$

$$a_i := a_i + \frac{j-1}{N}(b-a)$$

$$b_i := a_i + \frac{j}{N}(b-a)$$

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By (...) f is uniformly continuous so that for every $\varepsilon > 0$ there exists $\delta > 0$ such that

$$|x - y| < \delta \implies \|f(x) - f(y)\| < \varepsilon$$

Choose N such that

$$\max_i \frac{b_i - a_i}{N} < \delta$$

For every $x \in [a, b]$ we have $x \in [a_i, b_i)$ for some i and by construction $\|f(x) - f(a_i)\| < \varepsilon$ whence $\|f(x) - f_n(x)\| < \varepsilon$. As this applies for any $x \in [a, b]$ we see that the convergence is uniform. ∎

Proposition 5.8.17 (Integration of Regulated Functions)
Let X be a Banach space and $f : [a, b] \to X$ a function for which $f_n \to f$ converges uniformly. Then the sequence

$$\int_a^b f_n$$

converges in X. Furthermore the limit is independent of the choice of f_n, and we denote the limit by $\int_a^b f$.

In particular the integration of continuous functions with compact support is well-defined.

Proof. Let $x_n := \int f_n$. Then by (...) we have

$$\|x_n - x_m\| \leq (b - a)\|f_n - f_m\|_\infty$$

By the uniform convergence of f_n we see that x_n is a cauchy sequence, and so by assumption $x_n \to x$. The uniqueness is straightforward. ∎

Lemma 5.8.18
Let X be a Banach space and $f \in \mathrm{Reg}([a, b], X)$. Then $\|f\| \in \mathrm{Reg}([a, b], \mathbb{R})$.

Proposition 5.8.19 (Properties of the Integral)
Let X be a Banach space. Then the following properties hold

a) $\int_a^b (\lambda f + \mu g) = \lambda \int_a^b f + \mu \int_a^b g$
b) $\int_a^b f + \int_b^c f = \int_a^c f$
c) $\left\| \int_a^b f \right\| \leq \int_a^b \|f\|$

In the case $X = \mathbb{R}$ then we also have the following properties

d) $f \geq 0 \implies \int_a^b f \geq 0$
e) $f \leq g \implies \int_a^b f \leq \int_a^b g$
f) f continuous, positive such that $\int_a^b f = 0$ implies $f \equiv 0$

Proposition 5.8.20 (Fundamental Theorem of Calculus)
Let $f \in \mathrm{Reg}([a, b], X)$ where X is a Banach space. Suppose $c \in (a, b)$ is a point where f is continuous. Define

$$F(x) := \int_a^x f$$

Then F is differentiable at c and $F'(c) = f(c)$

Proof. Observe that for $0 < h < b - c$

$$\begin{aligned} \left\| \frac{F(c + h) - F(c)}{h} - f(c) \right\| &= \frac{1}{h} \left\| \int_c^{c+h} (f(t) - f(c))\, \mathrm{d}t \right\| \\ &\leq \frac{1}{h} \int_c^{c+h} \|f(t) - f(c)\|\, \mathrm{d}t \\ &\leq \sup_{|t-c|<h} \|f(t) - f(c)\| \end{aligned}$$

and similarly for $a - c < h < 0$. As f is regulated it is bounded (...) and so the supremum is finite. As f is assumed continuous we find $F'(c) = f(c)$. ∎

Proposition 5.8.21
Suppose $f_i \in \mathrm{Reg}([a, b], X_i)$ for $i = 1...n$. Then $(f_1, ..., f_n) \in \mathrm{Reg}([a, b], X)$ where $X := X_1 \times ...X_n$ and

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$$\int_a^b f = \left(\int_a^b f_1, ..., \int_a^b f_n \right)$$

5.8.4 Measure Spaces

To create a robust theory of integration it is necessary to replace finite additivity with countably additivity, and further show that such set functions can be defined over σ-algebras and not just algebras.

Definition 5.8.22 (Countably Additive Set Function)
Let $\mathcal{F}$ be a family of subsets of Ω containing $\emptyset$. A function

$$\mu : \mathcal{F} \to [0, \infty]$$

*is a **countably additive set function** (or **measure**) if it satisfies*

a) $\mu(\emptyset) = 0$
b) $\mu\left(\bigsqcup_{n=1}^{\infty} A_n\right) = \sum_{i=1}^{\infty} \mu(A_i)$ whenever this is well-defined

*We say it is **countably subadditive** if it satisfies*

$$\mu\left(\bigcup_{i=1}^{\infty} A_i \right) \leq \sum_{i=1}^{\infty} \mu(A_i)$$

whenever this is well-defined. We show in (5.8.25) that these conditions are almost equivalent.

Definition 5.8.23 (Measurable Space)
*Let $(\Omega, \mathcal{F})$ be a measurable space and $\mu : \mathcal{F} \to [0, \infty]$ a measure. Then the triplet $(\Omega, \mathcal{F}, \mu)$ is a **measurable space**.*

The following are standard results which are somewhat more awkward to prove in the case of semi-rings. However the proofs follow precisely the same lines and semi-ring extension theorem may be more directly applicable.

Proposition 5.8.24 (Continuity of Measures)
Let $\mathcal{R}$ be a semi-ring of subsets of Ω and $\mu : \mathcal{R} \to [0, \infty]$ a countably additive measure. Then for any sequence of sets $A_1, A_2, ... \in \mathcal{R}$ we have

$$\lim_{n \to \infty} \mu\left(\bigcup_{i=1}^{n} A_i \right) = \mu\left(\bigcup_{n=1}^{\infty} A_n \right)$$

whenever this is well-defined.

Proof. Define the increasing sequence

$$B_n := \bigcup_{i=1}^{n} A_i$$

then wish to show

$$\lim_{n \to \infty} \mu(B_n) = \mu\left(\bigcup_{n=1}^{\infty} B_n \right)$$

By definition we have (with the convention $B_0 = \emptyset$)

$$B_i \setminus \bigcup_{j=1}^{i-1} B_j = B_i \setminus B_{i-1} = \bigsqcup_{k=1}^{N_i} C_{ik} \quad i = 1, 2, ...$$

for $C_{ik} \in \mathcal{R}$ whence

$$B_n = \bigsqcup_{i=1}^{n} B_i \setminus B_{i-1} = \bigsqcup_{i=1}^{n} \bigsqcup_{k=1}^{N_n} C_{ik}$$

for $n = 1, 2, ..., \infty$. Then by additivity

$$\lim_{n \to \infty} \mu(B_n) = \lim_{n \to \infty} \sum_{i=1}^{n} \sum_{k=1}^{N_i} \mu(C_{ik}) = \sum_{i=1}^{\infty} \sum_{k=1}^{N_i} \mu(C_{ik})$$

and

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$$\mu\left(\bigcup_{n=1}^{\infty} B_n\right) = \mu\left(\bigsqcup_{i=1}^{\infty}\bigsqcup_{k=1}^{N_i} C_{ik}\right) = \sum_{i=1}^{\infty}\sum_{k=1}^{N_i}\mu(C_{ik})$$

■

Proposition 5.8.25 (Finitely Additive + Countably Subadditive $\iff$ Countably Additive)
Let $\mathcal{R}$ be a semi-ring of subsets of Ω and $\mu : \mathcal{R} \to [0,\infty]$ a finitely additive set function. Then

$$\mu\left(\bigsqcup_{i=1}^{\infty} A_i\right) \geq \sum_{i=1}^{\infty}\mu(A_i)$$

whenever this is well-defined. In particular μ is a measure iff it is countably subadditive.

Proof. By (5.8.9).b) we have

$$\sum_{i=1}^{n}\mu(A_i) \leq \mu\left(\bigsqcup_{i=1}^{\infty} A_i\right)$$

and the result follows by taking $n \to \infty$ (5.1.21).

Therefore if μ is countably subadditive it must also be countably additive. Conversely if μ is countably additive then we aim to show

$$\mu\left(\bigcup_{i=1}^{\infty} A_i\right) \leq \sum_{i=1}^{\infty}\mu(A_i)$$

for a not necessarily disjoint sequence $A_i \in \mathcal{R}$. By assumption μ is finitely additive, so by (5.8.9).c)

$$\mu\left(\bigcup_{i=1}^{n} A_i\right) \leq \sum_{i=1}^{n}\mu(A_i) \leq \sum_{i=1}^{\infty}\mu(A_i)$$

Using continuity of measures (5.8.24) we deduce that

$$\mu\left(\bigcup_{i=1}^{\infty} A_i\right) = \lim_{n\to\infty}\mu\left(\bigcup_{i=1}^{n} A_i\right) \leq \sum_{i=1}^{\infty}\mu(A_i)$$

as required. ■

Proposition 5.8.26 (Extension Theorem II)
Let $\mathcal{R}$ be a semi-ring of sets over Ω and $\mu : \mathcal{R} \to [0,\infty]$ a measure. Suppose that Ω is denumerable union of sets in $\mathcal{R}$. Then there exists an extension

$$\tilde{\mu} : \sigma(\mathcal{R}) \to [0,\infty]$$

to the σ-algebra generated by $\mathcal{R}$. When μ is σ-finite then this extension is unique. More precisely it is given by

$$\tilde{\mu}(A) := \inf\left\{\sum_{i=1}^{\infty}\mu(A_i) \;\middle|\; A \subset \bigcup_{i=1}^{\infty} A_i, A_i \in \mathcal{R}\right\}$$

Proof. Define first the "outer measure"

$$\mu^\star : \mathcal{P}(\Omega) \to [0,\infty]$$
$$\mu^\star(A) := \inf\left\{\sum_{i=1}^{\infty}\mu(E_i) \;\middle|\; A \subset \bigcup_{i=1}^{\infty} E_i \quad E_i \in \mathcal{F}\right\}$$

Note that the infimum is well-defined precisely by the hypothesis. Further define the family of subsets

$$\hat{\mathcal{R}} := \{A \subset \Omega \mid \mu^\star(B) = \mu^\star(A\cap B) + \mu^\star(A^c \cap B) \quad \forall B \subset \Omega\}$$

We make a number of claims

a) $\mu^\star(A) = \mu(A)$ for all $A \in \mathcal{R}$
b) $\mu^\star$ is countably subadditive and monotone

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$$\mu^\star\left(\bigcup_{i=1}^{\infty} A_i\right) \leq \sum_{i=1}^{\infty} \mu^\star(A_i) \quad A_i \subset \Omega$$

c) $\mathcal{R} \subset \hat{\mathcal{R}}$

d) $\hat{\mathcal{R}}$ is a σ-algebra containing $\sigma(\mathcal{R})$ and $\mu^\star$ is a measure on $\hat{\mathcal{R}}$

and prove each in turn

a) Evidently $\mu^\star(A) \leq \mu(A)$ as $A \subset A \cup \emptyset \ldots \cup \emptyset \ldots$. Conversely suppose that $A \subset \bigcup_{i=1}^{\infty} A_i$ for $A_i \in \mathcal{R}$ and $\mu^\star(A) < \infty$ (otherwise the reverse inequality is clear). Then evidently

$$A = \bigcup_{i=1}^{\infty} (A_i \cap A)$$

As μ is countably subadditive (5.8.25) and monotone (5.8.9) we see

$$\mu(A) \leq \sum_{i=1}^{\infty} \mu(A_i \cap A) \leq \sum_{i=1}^{\infty} \mu(A_i)$$

By definition of the infimum we find $\mu(A) \leq \mu^\star(A)$.

b) Let $A_1, \ldots, A_n, \ldots$ be subsets of Ω such that $A \subset \bigcup_{i=1}^{\infty} A_i$. For a given $\varepsilon > 0$, choose $A_{ij} \in \mathcal{R}$ such that $A_i \subset \bigcup_{j=1}^{\infty} A_{ij}$ and

$$\sum_{j=1}^{\infty} \mu(A_{ij}) \leq \mu^\star(A_i) + \frac{\varepsilon}{2^i}$$

Then evidently $A \subset \bigcup_{i,j=1}^{\infty} A_{ij}$ and so by definition

$$\mu^\star(A) \leq \sum_{i,j=1}^{\infty} \mu(A_{ij}) \leq \sum_{i=1}^{\infty} \mu^\star(A_i) + \varepsilon$$

As ε was arbitrary we deduce that $\mu^\star$ is countably subadditive. Monotonicity is straightforward from the definition.

c) Suppose $A \in \mathcal{R}$ and $B \subset \Omega$ then we require to prove that

$$\mu^\star(B) = \mu^\star(A \cap B) + \mu^\star(A^c \cap B)$$

By subadditivity it is sufficient so show that

$$\mu^\star(B) \geq \mu^\star(A \cap B) + \mu^\star(A^c \cap B)$$

Let $B \subset \bigcup_{i=1}^{\infty} B_i$ be an arbitrary cover with $B_i \in \mathcal{R}$, then by definition of $\mu^\star$ it is sufficient to show that

$$\mu^\star(A \cap B) + \mu^\star(A^c \cap B) \leq \sum_{i=1}^{\infty} \mu(B_i)$$

Recall by definition of semi-ring that

$$A^c \cap B_i = \bigsqcup_{k=1}^{n_i} C_{ik}$$

for some disjoint sets $C_{ik} \in \mathcal{R}$ and evidently

$$A \cap B \subset \bigcup_{i=1}^{\infty} (A \cap B_i)$$

$$A^c \cap B \subset \bigcup_{i=1}^{\infty} \bigsqcup_{k=1}^{n_i} C_{ik}$$

Therefore by monotonicity and countable subadditivity of $\mu^\star$

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$$\begin{aligned}\mu^\star(A\cap B)+\mu^\star(A^c\cap B) &\leq \mu^\star\left(\bigcup_{i=1}^{\infty}(A\cap B_i)\right)+\mu^\star\left(\bigcup_{i=1}^{\infty}\bigsqcup_{k=1}^{n_i}C_{ik}\right)\\ &\leq \sum_{i=1}^{\infty}\mu^\star(A\cap B_i)+\sum_{i=1}^{\infty}\sum_{k=1}^{n_i}\mu(C_{ik})\\ &= \sum_{i=1}^{\infty}\mu(A\cap B_i)+\sum_{i=1}^{\infty}\sum_{k=1}^{n_i}\mu(C_{ik})\\ &= \sum_{i=1}^{\infty}\left(\mu(A\cap B_i)+\sum_{k=1}^{n_i}\mu(C_{ik})\right)\\ &= \sum_{i=1}^{\infty}\mu(B_i)\end{aligned}$$

where the last step follows from finite additivity of μ.

d) We prove the various properties in turn

i) Evidently $\mu^\star(\emptyset)=0$ and therefore $\emptyset,\Omega\in\hat{\mathcal{R}}$

ii) Given $A_1,A_2\in\hat{\mathcal{R}}$ and $B\subset\Omega$. Observe that we may apply the measurability condition to B and A_1, and then to $B\cap A_1$ and A_2, and $B\cap A_1^c$ and A_2 to find

$$\begin{aligned}\mu^\star(B) &= \mu^\star(B\cap A_1)+\mu^\star(B\cap A_1^c)\\ &= \mu^\star(B\cap A_1\cap A_2)+\mu^\star(B\cap A_1\cap A_2^c)+\mu^\star(B\cap A_1^c\cap A_2)+\mu^\star(B\cap A_1^c\cap A_2^c)\end{aligned}$$

We may also consider the relationship with B replaced with $B\cap(A_1\cup A_2)$ to find

$$\mu^\star(B\cap(A_1\cup A_2))=\mu^\star(B\cap A_1\cap A_2)+\mu^\star(B\cap A_1\cap A_2^c)+\mu^\star(B\cap A_1^c\cap A_2) \qquad (*)$$

Combining these two relations we deduce

$$\mu^\star(B)=\mu^\star(B\cap(A_1\cup A_2))+\mu^\star(B\cap(A_1\cup A_2)^c)$$

As B was arbitrary then we deduce $A_1\cup A_2\in\hat{\mathcal{R}}$. Evidently by definition $A^c\in\hat{\mathcal{R}}$ and therefore by de-morgan $A_1\cap A_2\in\hat{\mathcal{R}}$. Therefore $\hat{\mathcal{R}}$ is closed under finite unions and intersections.

iii) Let $A_1,...,A_n\in\hat{\mathcal{R}}$ be disjoint then we claim that

$$\mu^\star\left(B\cap\left(\bigsqcup_{i=1}^{n}A_i\right)\right)=\sum_{i=1}^{n}\mu^\star(B\cap A_i)$$

From (*) we have

$$\mu^\star(B\cap(A_1\sqcup A_2))=\mu^\star(B\cap A_1)+\mu^\star(B\cap A_2)$$

and the result follows by induction.

iv) Let $A_1,A_2,...\in\hat{\mathcal{R}}$ be a countable family and $B\subset\Omega$. We require to prove that

$$\mu^\star(B)=\mu^\star\left(B\cap\bigcup_{i=1}^{\infty}A_i\right)+\mu^\star\left(B\cap\left(\bigcup_{i=1}^{\infty}A_i\right)^c\right)$$

By (5.8.6) we may consider the decomposition

$$A_i\setminus\bigcup_{j=1}^{i-1}A_j=\bigsqcup_{k=1}^{n_i}C_{ik}$$

where $C_{ik}\in\mathcal{R}$ are disjoint. Observe

$$\bigcup_{i=1}^{n}A_i=\bigsqcup_{i=1}^{n}\bigsqcup_{k=1}^{n_i}C_{ik}\quad\text{for } n=1,2,...,\infty$$

We have shown in ii) that $\bigcup_{i=1}^{n}A_i\in\hat{\mathcal{R}}$, therefore

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$$\begin{aligned}\mu^\star(B) &= \mu^\star\left(B \cap \left(\bigcup_{i=1}^{n} A_i\right)\right) + \mu^\star\left(B \cap \left(\bigcup_{i=1}^{n} A_i\right)^c\right) \\ &= \mu^\star\left(B \cap \left(\bigsqcup_{i=1}^{n}\bigsqcup_{k=1}^{n_i} C_{ik}\right)\right) + \mu^\star\left(B \cap \left(\bigcup_{i=1}^{n} A_i\right)^c\right)\end{aligned}$$

Recall from iii) that $\mu^\star(B \cap -)$ is finitely additive on $\hat{\mathcal{R}}$. Additionally by monotonicity we may deduce that

$$\begin{aligned}\mu^\star(B) &= \sum_{i=1}^{n}\sum_{k=1}^{n_i} \mu^\star(B \cap C_{ik}) + \mu^\star\left(B \cap \left(\bigcup_{i=1}^{n} A_i\right)^c\right) \\ &\geq \sum_{i=1}^{n}\sum_{k=1}^{n_i} \mu^\star(B \cap C_{ik}) + \mu^\star\left(B \cap \left(\bigcup_{i=1}^{\infty} A_i\right)^c\right)\end{aligned}$$

Letting $n \to \infty$ we find by subadditivity

$$\begin{aligned}\mu^\star(B) &\geq \sum_{i=1}^{\infty}\sum_{k=1}^{n_i} \mu^\star(B \cap C_{ik}) + \mu^\star\left(B \cap \left(\bigcup_{i=1}^{\infty} A_i\right)^c\right) \\ &\geq \mu^\star\left(B \cap \left(\bigcup_{i=1}^{\infty} A_i\right)\right) + \mu^\star\left(B \cap \left(\bigcup_{i=1}^{\infty} A_i\right)^c\right)\end{aligned}$$

However by subadditivity the reverse inequality holds and we deduce that these are all equal. This shows that $\bigcup_{i=1}^{\infty} A_i \in \hat{\mathcal{R}}$ and $\hat{\mathcal{R}}$ is closed under countable unions. Consider further the case A_i are disjoint and $B \coloneqq \bigsqcup_{j=1}^{\infty} A_j$. Then we may assume that $n_i = 1$ and $C_{i1} = A_i$. Using the above relationship

$$\mu^\star\left(\bigsqcup_{j=1}^{\infty} A_j\right) = \sum_{i=1}^{\infty} \mu^\star\left(\left(\bigsqcup_{j=1}^{\infty} A_j\right) \cap A_i\right) = \sum_{i=1}^{\infty} \mu^\star(A_i)$$

This shows that $\mu^\star$ is countably additive.

■

Definition 5.8.27 (Complete Set Function)
*Let $(\Omega, \mathcal{F}, \mu)$ be a measure space. We say it is **complete** if for every null set N (such that $\mu(N) = 0$) and $N' \subseteq N$ we have $N' \in \mathcal{F}$.*

Proposition 5.8.28 (Completion of a Measure Space)
*Let $(\Omega, \mathcal{F}, \mu)$ be a measure space and define the **completion** by*

$$\begin{aligned}\overline{\mathcal{F}} &\coloneqq \{A \cup Z \mid A \in \mathcal{F}, Z \subseteq N, \mu(N) = 0\} \\ \overline{\mu}(A \cup Z) &\coloneqq \mu(A)\end{aligned}$$

Then $\left(\Omega, \overline{\mathcal{F}}, \overline{\mu}\right)$ is a well-defined complete measure space. Furthermore $\overline{\mathcal{F}}$ is the smallest complete σ-algebra containing $\mathcal{F}$, and $\overline{\mu}$ is the unique extension.

5.8.5 Borel Measure on $\mathbb{R}$

Proposition 5.8.29
Let $\mathcal{R}$ be the semi-ring of half-open intervals over $\mathbb{R}$. Then the set function

$$\mu : \mathcal{R} \to [0, \infty]$$

given by

$$\mu([a, b)) = b - a,$$

is countably additive and σ-finite.

Proof. Evidently μ is finitely additive, therefore by (5.8.25) it is sufficient to show that it is countably subadditive. Suppose that

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$$\bigcup_{i=1}^{\infty}[a_i, b_i) = [a, b)$$

Let $\varepsilon < b - a$. Then $[a, b-\varepsilon]$ is compact by (5.7.7) and $U_i := \left(a_i - \frac{\delta}{2^i}, b_i\right)$ is open for every $\delta > 0$. By definition of compactness there is some finite integer n such that

$$\left[a, \tilde{b}\right] := [a, b-\varepsilon] \subset \bigcup_{i=1}^{n}\left(a_i - \frac{\delta}{2^i}, b_i\right) =: (\tilde{a}_i, b_i)$$

By reordering we assume that $\tilde{a}_1 < ... < \tilde{a}_n$. As $\left[a, \tilde{b}\right]$ is an interval we must have $\tilde{a}_{i+1} < b_i$. Further we may assume that $b_i < b_{i+1}$ (otherwise we may omit the interval $(\tilde{a}_i, b_i)$). By construction we must also have $\tilde{a}_1 < a$ and $\tilde{b} < b_n$.

Consequently

$$b - a - \varepsilon = \tilde{b} - a \leq b_n - \tilde{a}_1 = b_1 - \tilde{a}_1 + \sum_{i=1}^{n-1}(b_{i+1} - b_i) \leq \sum_{i=1}^{n}(b_i - \tilde{a}_i) \leq \sum_{i=1}^{n}(b_i - a_i) + \delta \leq \sum_{i=1}^{\infty}(b_i - a_i) + \delta$$

This shows

$$\mu([a, b)) = \mu\left(\bigcup_{i=1}^{\infty}[a_i, b_i)\right) \leq \sum_{i=1}^{\infty}\mu([a_i, b_i))$$

as required. ∎

Proposition 5.8.30
Let $D \subset \mathbb{R}$ be an additive subgroup containing $\mathbb{Q}$. Then $\mathcal{B}(\mathbb{R})$ is generated by any of the following families of intervals

1. $(a, b]$ $a < b \in D$
2. (a, b) $a < b \in D$
3. $[a, b)$ $a < b \in D$
4. (a, ∞) $a \in D$
5. $[a, \infty)$ $a \in D$
6. $(-\infty, a]$ $a \in D$
7. $(-\infty, a)$ $a \in D$

Further there is a unique measure

$$\mu : \mathcal{B}(\mathbb{R}) \to [0, \infty]$$

such that

$$\mu([a, b)) = b - a$$

Proof. Denote the corresponding σ-algebras by $\mathcal{F}_1, ..., \mathcal{F}_7$. Observe that by the Archimedean property

$$(a, b) = \bigcup_{n=1}^{\infty}\left(a, b - \frac{1}{n}\right] = \bigcup_{n=1}^{\infty}\left[a + \frac{1}{n}, b\right)$$

$$[a, b) = \bigcap_{n=1}^{\infty}\left(a - \frac{1}{n}, b\right)$$

$$(a, b] = \bigcap_{n=1}^{\infty}\left(a, b + \frac{1}{n}\right)$$

so that $\mathcal{F}_1 = \mathcal{F}_2 = \mathcal{F}_3$. We may argue similarly $\mathcal{F}_4 = \mathcal{F}_5 = \mathcal{F}_6 = \mathcal{F}_7$. Finally

$$(a, b) = (a, \infty) \cap (-\infty, b)$$

and

$$(a, \infty) = \bigcup_{n=1}^{\infty}(a, n)$$

to show these are equal. The measure μ exists by (5.8.29) and (5.8.26). ∎

Proposition 5.8.31 (Borel Measure on the Extended Real Line)
Let $\mathcal{B}(\mathbb{R}^\sharp)$ be the Borel Measure on the extended real line. Then $\mathcal{B}(\mathbb{R}^\sharp)$ consists of sets of the form

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$$\left\{A \subset \mathbb{R}^\sharp \mid A \cap \mathbb{R} \in \mathcal{B}(\mathbb{R})\right\}$$

or equivalently

$$\{A \cup B \mid A \in \mathcal{B}(\mathbb{R}), B \in \mathcal{P}(\{-\infty, \infty\})\}$$

Furthermore it is generated by any of the following families

1. $(a, \infty]$
2. $[a, \infty]$
3. $[-\infty, a)$
4. $[-\infty, a]$

for $a \in D$ where $D \subset \mathbb{R}$ is an additive subgroup containing $\mathbb{Q}$.

Proof. If $U \subset \mathbb{R}$ is open, then it is also open in $\mathbb{R}^\sharp$. Therefore $\mathcal{B}(\mathbb{R}) \subset \mathcal{B}(\mathbb{R}^\sharp)$. Evidently $\mathbb{R}$, $\{\infty\}$, $\{-\infty\} \in \mathcal{B}(\mathbb{R}^\sharp)$ as they are open and closed respectively. This shows that the given family equals $\mathcal{B}(\mathbb{R}^\sharp)$.

Denote by $\mathcal{F}_1, ..., \mathcal{F}_4$ the σ-algebras generated by the respective families of rays. Note that $\mathcal{F}_1 = \mathcal{F}_4$ and $\mathcal{F}_2 = \mathcal{F}_3$ by taking complements. Further $\{\infty\} \in \mathcal{F}_1 \cap \mathcal{F}_2$ by taking intersections as $a \to \infty$ and similarly $\{-\infty\} \in \mathcal{F}_4 \cap \mathcal{F}_3$. This shows that they also contain either the open rays (a, ∞) or the semi-open rays $[a, \infty)$. Therefore they contain $\mathcal{B}(\mathbb{R})$ by (5.8.30). Then by the characterisation in the first part we see that this equals $\mathcal{B}(\mathbb{R}^\sharp)$. ∎

5.8.6 Product Measure

Definition 5.8.32 (Product of σ-algebra)
*Let $(\Omega_1, \mathcal{F}_1), ..., (\Omega_n, \mathcal{F}_n)$ be a family of measurable spaces. Then define the **product measurable space** $(\Omega_1 \times ... \times \Omega_n, \mathcal{F}_1 \otimes ... \otimes \mathcal{F}_n)$ to be given by*

$$\mathcal{F}_1 \otimes ... \otimes \mathcal{F}_n := \sigma(\mathcal{F}_1 \times ... \times \mathcal{F}_n)$$

and

$$\mathcal{F}_1 \times ... \times \mathcal{F}_n := \{A_1 \times ... \times A_n \mid A_i \in \mathcal{F}_i\}$$

Definition 5.8.33 (Product Measure)
Let $(\Omega_i, \mathcal{F}_i, \mu_i)$ be a family of σ-finite measures for $i = 1...n$. Then we say a measure

$$\mu_1 \otimes ... \otimes \mu_n : \mathcal{F}_1 \otimes ... \otimes \mathcal{F}_n \to \mathbb{R}^\sharp$$

such that

$$\mu(A_1 \times ... \times A_n) = \mu_1(A_1) \cdot ... \cdot \mu_{n(A_n)}$$

*is called the **product measure space**.*

We defer proof of existence and uniqueness as it relies on theory of integration, but nevertheless prove the finitely additive case and then the case of $\mathbb{R}^n$ in the next section.

Lemma 5.8.34
Let $\mathcal{R}_1, ..., \mathcal{R}_n$ be semi-rings. Then the family

$$\mathcal{R}_1 \times ... \times \mathcal{R}_n = \{A_1 \times ... \times A_n \mid A_i \in \mathcal{R}_i\}$$

is a semi-ring.

Proof. We may reduce to the case of $n = 2$. Then observe that

$$(A_1 \times B_1) \setminus (A_2 \times B_2) = (A_1 \setminus A_2) \times (B_1 \setminus B_2) \sqcup (A_1 \cap A_2) \times (B_1 \setminus B_2) \sqcup (A_1 \setminus A_2) \times (B_1 \cap B_2)$$

∎

Lemma 5.8.35
Let $\mathcal{R}$ be a semi-ring and $A_1, ..., A_n \in \mathcal{R}$ a family of subsets. Then there exists a family of disjoint subsets $\mathcal{S} := \{B_1, ..., B_N\} \subset \mathcal{R}$ such that $\mathcal{S} = \mathcal{S}_1 \cup ... \cup \mathcal{S}_n$ and

$$A_i = \sqcup_{B \in \mathcal{S}_i} B$$

Proof. For a tuple $\varepsilon := (\varepsilon_1, ..., \varepsilon_n)$ for which $\varepsilon_i = \pm 1$ define the family of disjoint sets

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$$A^\varepsilon := A_1^{\varepsilon_1} \cap ... \cap A_n^{\varepsilon_n}$$

where $A_i^1 := A_i$ and $A_i^{-1} := A_i^c$. By (5.8.6) A^ε may be expressed as the disjoint union of some sets in $\mathcal{R}$, say $\mathcal{S}^\varepsilon$. Then $\mathcal{S} = \cup_\varepsilon \mathcal{S}^\varepsilon$ consists of pairwise disjoint sets, where we consider tuples ε for which at least one entry is positive. Further

$$A_i = \sqcup_{\varepsilon:\varepsilon_i=1} A^\varepsilon$$

therefore we may define $\mathcal{S}_i := \cup_{\varepsilon:\varepsilon_i=1} \mathcal{S}^\varepsilon$. Note throughout we may omit ε for which $A^\varepsilon = \emptyset$. ■

Lemma 5.8.36
Let $\{(\Omega_i, \mathcal{R}_i, \mu_i)\}_{i=1...n}$ be a collection of finitely additive set functions on semi-rings. The set function

$$\mu : \mathcal{R}_1 \times ... \times \mathcal{R}_n \to \mathbb{R} \cup \{\infty\}$$
$$A_1 \times ... \times A_n \to \mu_1(A_1)...\mu_{n(A_n)}$$

is finitely additive, and we write $\mu =: \mu_1 \times ... \times \mu_n$. Moreover if each μ_i is σ-finite then so is μ.

Proof. By induction it's enough to consider the case $n = 2$. Suppose that

$$A_i = B_i \times C_i \quad i = 0, ..., n$$
$$A_0 = \sqcup_{i=1}^n A_i$$

for $B_i \in \Omega_1$ and $C_i \in \Omega_2$. By (5.8.35) we may write

$$B_i := \sqcup_{\hat{B} \in \mathcal{S}_i} \hat{B}$$
$$C_i := \sqcup_{\hat{C} \in \mathcal{T}_i} \hat{C}$$

where $\mathcal{S}_i$ and $\mathcal{T}_i$ are finite subsets of $\mathcal{R}_1$ and $\mathcal{R}_2$ respectively. Evidently

$$B_i \times C_i = \sqcup_{(\hat{B},\hat{C}) \in \mathcal{S}_i \times \mathcal{T}_i} \hat{B} \times \hat{C}$$

and $\mathcal{S}_0 = \cup_{i=1}^n \mathcal{S}_i$ and $\mathcal{T}_0 := \cup_{i=1}^n \mathcal{T}_i$. As μ_1 and μ_2 are finitely additive then

$$\mu_1(B_i) = \sum_{\hat{B} \in \mathcal{S}_i} \mu_1\big(\hat{B}\big)$$
$$\mu_2(C_i) = \sum_{\hat{C} \in \mathcal{T}_i} \mu_2\big(\hat{C}\big)$$
$$\sum_{i=1}^n \mu(A_i) = \sum_{i=1}^n \mu_1(B_i)\mu_2(C_i) = \sum_{i=1}^n \sum_{(\hat{B},\hat{C}) \in \mathcal{S}_i \times \mathcal{T}_i} \mu_1\big(\hat{B}\big)\mu_2\big(\hat{C}\big)$$
$$\mu(A_0) = \mu_1(B_0)\mu_2(C_0) = \sum_{(\hat{B},\hat{C}) \in \mathcal{S}_0 \times \mathcal{T}_0} \mu_1\big(\hat{B}\big)\mu_2\big(\hat{C}\big)$$

As the sets $B_i \times C_i$ are disjoint we may deduce that

$$\mathcal{S}_0 \times \mathcal{T}_0 = \sqcup_{i=1}^n \mathcal{S}_i \times \mathcal{T}_i$$

from which the result follows. ■

5.8.7 Borel Measure on $\mathbb{R}^n$

Proposition 5.8.37
Let $\mathcal{R}$ be the semi-ring of half-open intervals over $\mathbb{R}^n$. Then the set function

$$\mu : \mathcal{R} \to [0, \infty]$$

given by

$$\mu([a_1, b_1) \times ... \times [a_n, b_n)) = \prod_{i=1}^n (b_i - a_i),$$

is countably additive and σ-finite.

Proof. By (5.8.36) μ is finitely additive. So by (5.8.25) it is sufficient to show that μ is countably subadditive. Suppose that

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$$[\boldsymbol{a}_0, \boldsymbol{b}_0) = \bigcup_{i=1}^{\infty} [\boldsymbol{a}_i, \boldsymbol{b}_i)$$

Then for any positive constants $\boldsymbol{\varepsilon}$, $\boldsymbol{\delta}_1, \boldsymbol{\delta}_2, \ldots$ we have

$$[\boldsymbol{a}_0, \boldsymbol{b}_0 - \boldsymbol{\varepsilon}] \subset \bigcup_{i=1}^{\infty} (\boldsymbol{a}_i - \boldsymbol{\delta}_i, \boldsymbol{b}_i)$$

By the Heine-Borel theorem (5.7.7) closed intervals are compact, so we have

$$[\boldsymbol{a}_0, \boldsymbol{b}_0 - \boldsymbol{\varepsilon}) \subset \bigcup_{i=1}^{N(\boldsymbol{\delta})} [\boldsymbol{a}_i - \boldsymbol{\delta}_i, \boldsymbol{b}_i)$$

Therefore by (5.8.9)

$$\begin{aligned}
\prod_{j=1}^{n} (b_{0j} - a_{0j} - \varepsilon) &\leq \sum_{i=1}^{N(\boldsymbol{\delta})} \prod_{j=1}^{n} (b_{ij} - a_{ij} + \delta_{ij}) \\
&\leq \sum_{i=1}^{N(\boldsymbol{\delta})} \prod_{j=1}^{n} (b_{ij} - a_{ij}) \left(1 + \frac{\delta}{2^i}\right) \\
&\leq \sum_{i=1}^{N(\boldsymbol{\delta})} \prod_{j=1}^{n} (b_{ij} - a_{ij}) \sum_{i=1}^{N(\boldsymbol{\delta})} \left(1 + \frac{\delta}{2^i}\right) \\
&\leq (1 + \delta) \sum_{i=1}^{\infty} \prod_{j=1}^{n} (b_{ij} - a_{ij})
\end{aligned}$$

where we have defined $\delta_{ij} := \frac{b_{ij} - a_{ij}}{n} \left(\frac{\delta}{2^i}\right)^{\frac{1}{n-1}}$ and used the inequality $(1 + x)^n \leq 1 + nx^{n-1}$. As this holds for arbitrary $\delta > 0$ then it holds for $\delta = 0$. We may consider the sequence $\varepsilon := \frac{1}{m}$ and letting $m \to \infty$ use (4.1.50) and (5.1.21) to deduce the case of $\varepsilon = 0$. This shows that

$$\mu([\boldsymbol{a}_0, \boldsymbol{b}_0)) \leq \sum_{i=1}^{\infty} \mu([\boldsymbol{a}_i, \boldsymbol{b}_i))$$

i.e. μ is countably subadditive. ∎

Proposition 5.8.38 (Existence of Borel Measure)
Let $\left(\mathbb{R}^k, \mathcal{B}(\mathbb{R}^k)\right)$ be the Borel measurable space. Then there exists a unique σ-finite measure

$$\mu : \mathcal{B}(\mathbb{R}^k) \to [0, \infty]$$

such that

$$\mu([\boldsymbol{a}, \boldsymbol{b})) = \prod_{i=1}^{k} (b_i - a_i)$$

Proof. This is a consequence of (5.8.26) and (5.8.37). ∎

5.8.8 Measurable Functions

Definition 5.8.39
*Let $(\Omega_1, \mathcal{F}_1)$ and $(\Omega_2, \mathcal{F}_2)$ be measurable spaces. A function $f : \Omega_1 \to \Omega_2$ is said to be **measurable** if*

$$E \in \mathcal{F}_2 \implies f^{-1}(E) \in \mathcal{F}_1$$

We may write $\mathcal{F}_1/\mathcal{F}_2$-measurable in order to make the σ-algebras explicit.

Similarly a function $f : \Omega \to X$ from a measurable space to a topological space is said to be measurable if it is $\mathcal{F}/\mathcal{B}(X)$ measurable.

Proposition 5.8.40
The following properties hold

a) The composition of measurable functions is measurable
b) Suppose $\mathcal{F}_2 = \sigma(\mathcal{C})$ then a function $f : \Omega_1 \to \Omega_2$ is $\mathcal{F}_1/\mathcal{F}_2$-measurable iff

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$$E \in \mathcal{C} \implies f^{-1}(E) \in \mathcal{F}_1$$

c) A function $f : \Omega \to X$ is measurable precisely when the inverse image of an open set (resp. closed set) is measurable.

Corollary 5.8.41 (Criteria for measurablity of a real-valued function)
Let $(\Omega, \mathcal{F})$ be a measurable space and $f : \Omega \to \mathbb{R}^\sharp$ a function. Then the following are equivalent

a) f is $\mathcal{F}/\mathcal{B}(\mathbb{R}^\sharp)$-measurable
b) $f^{-1}((x, \infty]) \in \mathcal{F}$ for all $x \in D$
c) $f^{-1}([x, \infty]) \in \mathcal{F}$ for all $x \in D$
d) $f^{-1}([-\infty, x)) \in \mathcal{F}$ for all $x \in D$
e) $f^{-1}([-\infty, x]) \in \mathcal{F}$ for all $x \in D$

where $D \subset \mathbb{R}$ is an additive subgroup containing $\mathbb{Q}$.

Proof. We may combine (5.8.40).b) and (5.8.31). ∎

Proposition 5.8.42 (Measurable Space Isomorphism)
Let $(\Omega_1, \mathcal{F}_1)$ and $(\Omega_2, \mathcal{F}_2)$ be measurable spaces and $f : \Omega_1 \to \Omega_2$ a bijective function. Then the following are equivalent

a) f and f^{-1} are measurable
b) f induces a bijection $\mathcal{F}_1 \to \mathcal{F}_2$
c) f^{-1} induces a bijection $\mathcal{F}_2 \to \mathcal{F}_1$

*In this case we say that $f : (\Omega_1, \mathcal{F}_1) \xrightarrow{\sim} (\Omega_2, \mathcal{F}_2)$ is a **measurable space isomorphism**.*

Proposition 5.8.43 (Limits of Measurable Functions)
Let $f_n : (\Omega, \mathcal{F}) \to (\mathbb{R}^\sharp, \mathcal{B}(\mathbb{R}^\sharp))$ be a sequence of measurable functions. Define the functions $\sup f_n, \inf f_n, \limsup_n f_n, \liminf_n f_n$ as follows

$$(\sup f_n)(x) := \sup_n f_n(x)$$
$$(\inf f_n)(x) := \inf_n f_n(x)$$
$$\left(\limsup_n f_n\right)(x) := \limsup_n f_n(x)$$
$$\left(\liminf_n f_n\right)(x) := \liminf_n f_n(x)$$

Then these functions are $\mathcal{F}/\mathcal{B}(\mathbb{R}^\sharp)$-measurable.

Proof. Observe that

$$\left(\inf_n f_n\right)^{-1}([x, \infty]) = \bigcap_{n=1}^{\infty} f_n^{-1}([x, \infty])$$

so $\inf f_n$ is measurable by (5.8.41). Further $\sup_n f_n = -\inf_n(-f_n)$. ∎

Proposition 5.8.44
Let $f, g : (\Omega, \mathcal{F}) \to (\mathbb{R}^\sharp, \mathcal{B}(\mathbb{R}^\sharp))$ be measurable functions. Then the following functions are measurable

a) $f \pm g$
b) λf is measurable for $\lambda \in \mathbb{R}$
c) $f \cdot g$
d) $\max(f, g)$, $\min(f, g)$

Note when $f \pm g, fg$ is not well-defined (e.g. $\infty - \infty$ or $-\infty \times -\infty$) then we choose an arbitrary, but fixed, sentinel value.

Proof. We make use of (5.8.41) throughout.

a) Define $h(\omega) := f(\omega) + g(\omega)$ and let A be the set on which the sum is well-defined. Then it is evidently measurable and

$$\begin{aligned} h^{-1}((x, \infty]) &= \{\omega \in B_x^c \mid f(\omega) + g(\omega) > x\} \cup B_x \\ &= \{\omega \in A \mid f(\omega) + g(\omega) > x\} \cup B_x \\ &= \left(\bigcup_{r \in \mathbb{Q}} \{\omega \in A \mid f(\omega) > r\} \cap \{\omega \in A \mid g(\omega) > x - r\}\right) \cup B_x \end{aligned}$$

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where $B_x = \emptyset$ or A^c according to if the sentinel value of $f+g$ is $\leq x$. For the final equality the reverse inclusion is immediate. Conversely suppose $f(\omega)+g(\omega) > x$ then by (5.1.29) there exists some $r \in \mathbb{Q}$ such that $f(\omega) > r > x - g(\omega)$. The case $f-g$ follows immediately from b).

b) Observe that

$$(\lambda f)^{-1}((x,\infty]) = \begin{cases} \left(\frac{x}{\lambda}, \infty\right] & \lambda > 0 \\ \left[-\infty, \frac{x}{\lambda}\right) & \lambda < 0 \\ \Omega & \lambda = 0, x < 0 \\ \emptyset & \lambda = 0, x \geq 0 \end{cases}$$

c) First consider the case f^2 and observe

$$(f^2)^{-1}((x,\infty]) = \begin{cases} f^{-1}(\{-\infty,\infty\}) \cup f^{-1}\left(\left(\sqrt{x},\infty\right]\right) \cup f^{-1}\left(\left[-\infty,-\sqrt{x}\right)\right) & x \geq 0 \\ \Omega & x < 0 \end{cases}$$

Then we may deduce the general case because

$$(fg)(\omega) = \begin{cases} \frac{(f(\omega)+g(\omega))^2 - (f(\omega)-g(\omega))^2}{4} & f(\omega), g(\omega) \notin \{-\infty,\infty\} \\ \infty & (f(\omega), g(\omega)) \in \{(\infty,\infty),(-\infty,-\infty)\} \\ -\infty & (f(\omega), g(\omega)) \in \{(\infty,-\infty),(-\infty,\infty)\} \end{cases}$$

d) We may see this as a special case of (5.8.43)

■

5.8.9 Integration over a Measure

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